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Edge-AI System for Early Wound Infection Detection

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Abstract:

Wound infections are a major healthcare concern, often leading to delayed healing and serious complications if not detected early. Traditional monitoring methods rely on manual inspection, which is subjective and unsuitable for continuous observation, while advanced systems using bio-signals are complex, costly, and noise-prone.

To address these challenges, this project proposes a context-aware Edge-AI system for early wound infection detection using a non-invasive and low-cost multi-sensor approach. The system uses temperature and humidity sensors placed in a simulated wound environment and introduces a novel context-compensation method by calculating differential parameters such as ΔT (wound vs ambient temperature) and ΔH (humidity variation). This improves accuracy by reducing environmental effects.

An ESP32 microcontroller processes data locally using Edge-AI techniques such as filtering, trend analysis, and adaptive baseline learning to classify the wound condition as Normal, Warning, or Critical. The system enables real-time, efficient monitoring without cloud dependency.

The proposed solution is safe, scalable, and suitable for real-world healthcare applications, particularly in remote monitoring and wearable devices.

Introduction:

Wound care management is a critical aspect of healthcare, especially in cases involving chronic wounds, post-surgical recovery, and diabetic patients. One of the major challenges in wound management is the early detection of infection, as delayed identification can lead to severe complications such as tissue damage, prolonged healing time, or even life-threatening conditions. Traditionally, wound assessment is carried out through visual inspection and manual evaluation by healthcare professionals. However, this approach is subjective, intermittent, and highly dependent on clinical expertise.

With the advancement of technology, there has been growing interest in automated and continuous wound monitoring systems. Many existing solutions rely on bio-signals such as ECG or invasive sensing techniques, which increase system complexity, cost, and susceptibility to noise. These systems are often not suitable for real-time, low-cost, and home-based monitoring.

To overcome these limitations, this project introduces a context-aware, non-invasive wound monitoring system that uses multiple environmental sensors combined with Edge Artificial Intelligence (Edge-AI). The system measures wound temperature and humidity and compares them with ambient conditions to derive meaningful insights. By focusing on contextual parameters rather than absolute values, the system improves detection accuracy and reduces false alarms.

The integration of Edge-AI enables real-time data processing directly on the device, eliminating the need for continuous internet connectivity. This makes the system efficient, scalable, and suitable for wearable healthcare applications. The proposed solution is designed to be safe, low-cost, reliable, and practical, making it highly relevant for both clinical and home-care environments.

Problem Statement:

Early detection of wound infections is difficult due to unreliable and non-continuous monitoring methods. Manual inspection is inconsistent, while advanced systems are costly, complex, and unsuitable for continuous use. Environmental variations also reduce accuracy.

Therefore, a simple, non-invasive, and intelligent system is needed for continuous and accurate detection. This project proposes a context-aware Edge-AI solution using multi-sensor data fusion.

Objectives:

The main objectives of this project are:

Primary Objectives

- To design and develop a non-invasive system for early detection of wound infections
- To implement multi-sensor data acquisition using temperature and humidity sensors
- To apply context-aware analysis by calculating differential parameters such as ΔT (temperature difference) and ΔH (humidity variation)
- To develop an Edge-AI-based decision system for real-time classification of wound conditions

Secondary Objectives

- To reduce false alarms by incorporating ambient compensation techniques
- To implement intelligent features such as:
 - Moving average filtering
 - Trend analysis
 - Adaptive baseline learning
- To classify wound status into Normal, Warning, and Critical levels

Implementation Objectives

- To design a compact and efficient system using the ESP32 microcontroller
- To display real-time results using an OLED interface

- To create a safe and controlled demo setup using a simulated wound model

Future-Oriented Objectives

- To enable scalability for wearable healthcare devices
- To provide a foundation for integration with:
 - Mobile applications
 - Cloud-based monitoring
 - Advanced TinyML models

Hardware Components:

1. Microcontroller

- **ESP32**
 - Acts as the main processing unit
 - Handles sensor data acquisition and Edge-AI logic
 - Provides high processing power with low energy consumption
 - Supports future IoT integration (Wi-Fi/Bluetooth)

2. Sensors

a) Temperature Sensors

- **DS18B20 (×2)**
 - One sensor measures **wound temperature (T_w)**
 - One sensor measures **ambient temperature (T_a)**
 - Digital sensor with high accuracy
 - Used to calculate temperature difference (ΔT)

b) Humidity Sensor

- **DHT22 (×1)**
 - Measures humidity near the wound dressing
 - Detects moisture/exudate levels
 - Used to determine infection conditions

3. Display Unit

- **OLED Display (0.96" I2C)**

- Displays real-time data:
 - Temperature
 - Humidity
 - ΔT value
 - System status (Normal/Warning/Critical)

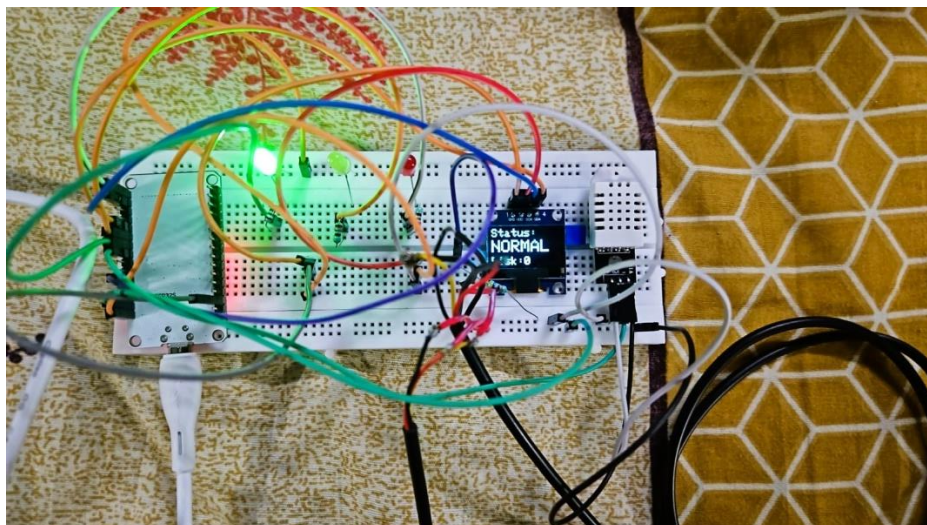
4. Supporting Components

- **Breadboard**
 - For circuit assembly and prototyping
- **Resistor (4.7k Ω & 220 Ω)**
 - Required for DS18B20 (pull-up resistor)
- **Jumper Wires**
 - For connections between components
- **USB Power Supply / Cable**
 - Provides power to ESP32

5. Optional Components (For Enhancement)

- **Mobile/Web Interface**
 - For visualization of sensor data
- **Battery Module**
 - For portable/wearable applications

Circuit:



Software / Algorithm Design:

The software component of the proposed system is responsible for acquiring sensor data, processing it using context-aware logic, and generating meaningful outputs. The system is programmed on the ESP32 microcontroller using embedded C/C++ (Arduino framework).

Data Acquisition

The ESP32 continuously reads input from:

- Two DS18B20 sensors (wound and ambient temperature)
- One DHT22 sensor (humidity)

The data is sampled at regular intervals to ensure real-time monitoring.

Feature Extraction

Instead of using raw sensor values, the system computes:

- **ΔT (Temperature Difference)** = Wound Temperature (T_w) – Ambient Temperature (T_a)
- **ΔH (Humidity Variation)** = Current Humidity – Baseline Humidity

This approach improves accuracy by compensating for environmental variations

Decision Logic

The system classifies wound condition into three levels:

- **Normal Condition**
 - $\Delta T < 1.0^{\circ}\text{C}$ AND Humidity $< 60\%$
- **Warning Condition**
 - ΔT between 1.0°C and 2.5°C
 - OR Humidity between 60% and 75%
- **Critical Condition**
 - $\Delta T \geq 2.5^{\circ}\text{C}$ AND Humidity $\geq 75\%$

Output Generation

Based on classification:

- Status is displayed on OLED
- Data is optionally logged or sent via serial communication

Edge AI Implementation:

The project incorporates lightweight Edge-AI techniques to enhance decision-making beyond simple threshold-based systems.

1. Moving Average Filtering

Sensor readings are smoothed using a moving average filter to:

- Reduce noise
- Eliminate sudden fluctuations
- Improve stability of readings

2. Trend Analysis

The system monitors trends in temperature and humidity:

- If ΔT increases continuously for 3 consecutive readings $\rightarrow$ risk level increases
- Detects gradual infection development

3. Adaptive Baseline Learning

- The system learns the **normal humidity baseline** over time
- Adjusts thresholds based on individual wound conditions
- Improves personalization and accuracy

4. Threshold Tuning

- Threshold values are dynamically adjusted
- Allows better sensitivity for different environments and patients

5. Edge Processing Advantages

- No need for cloud computation
- Faster response (real-time alerts)
- Improved data privacy
- Low power consumption

This makes the system intelligent, adaptive, and suitable for wearable healthcare applications

Working Principle:

The working principle of the system is based on **continuous monitoring, contextual analysis, and intelligent classification.**

Step-by-Step Operation

1. **Sensor Data Collection**
 - DS18B20 sensors measure wound and ambient temperature
 - DHT22 measures humidity near the wound dressing
2. **Data Processing**
 - ESP32 reads sensor values
 - Computes ΔT and ΔH
3. **Context-Aware Analysis**
 - Removes environmental influence using ambient compensation
 - Applies filtering and trend detection
4. **Decision Making**
 - Uses predefined thresholds and AI logic
 - Classifies condition as Normal, Warning, or Critical
5. **Output Display**
 - Results are shown on OLED display
 - Data can be monitored in real-time

Results and Observations:

The system was tested using a simulated wound environment to evaluate its performance under different conditions.

Test Conditions

1. Normal Condition

- Dry gauze
- Room temperature
- Output: **NORMAL**

2. Warning Condition

- Slight moisture applied
- Mild heat introduced
- Output: **WARNING**

3. Critical Condition

- High moisture level
- Increased temperature
- Output: **CRITICAL**

Observed Results

- The system successfully detected changes in temperature and humidity
- Context-aware parameters (ΔT , ΔH) reduced false alarms
- Smooth output achieved due to filtering
- Trend analysis helped identify gradual infection patterns

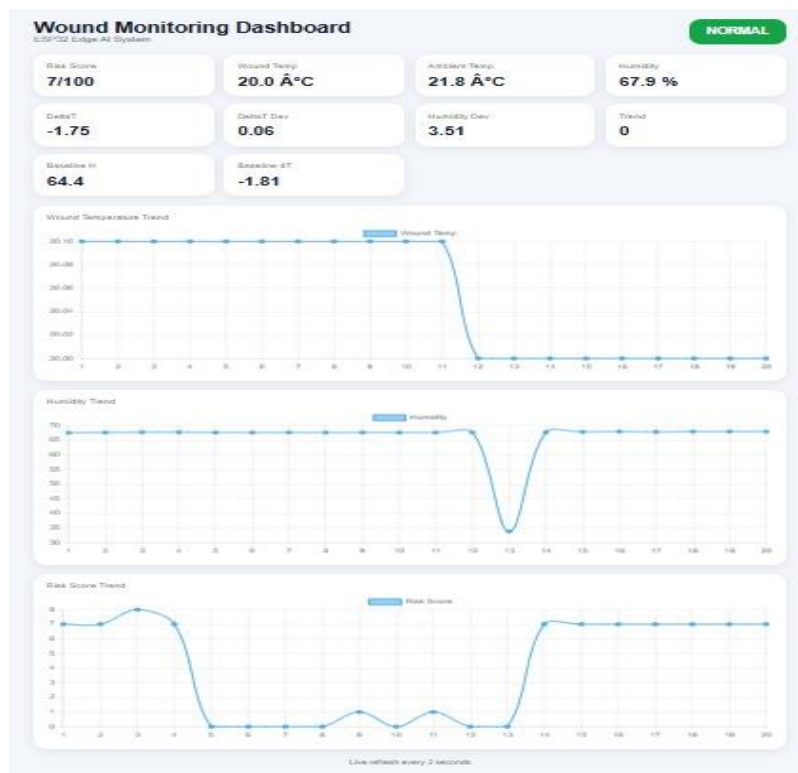
Performance Highlights

- Real-time response
- Stable and consistent readings
- Accurate classification across conditions

System Reliability

- No direct human contact required
- Safe and repeatable testing
- Suitable for practical deployment

Dashboard Output



Conclusion:

This project presents a context-aware Edge-AI system for early wound infection detection using temperature and humidity sensors. By using multi-sensor fusion and ambient compensation (ΔT and ΔH), the system improves accuracy and reduces false alarms without relying on complex bio-signals.

The ESP32 enables real-time, low-power, and efficient processing using lightweight Edge-AI techniques such as filtering and trend analysis. The system is non-invasive, low-cost, and suitable for clinical and home use.

Results from the simulated model show accurate classification of wound conditions, proving the system's reliability. Overall, the solution is simple, scalable, and has strong potential for future healthcare and wearable applications.

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